CBSE Practice Worksheet

Class 7 | Science | Difficulty: Hard | Set A

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. What is the wavelength of de Broglie wave for an electron?
   (A) $\lambda = h/mv$
   (B) $\lambda = mv/h$
   (C) $\lambda = hv/m$
   (D) $\lambda = m/hv$

2. What is the wavelength of Lyman series limit?
   (A) 91.2 nm
   (B) 121.6 nm
   (C) 656.3 nm
   (D) 364.6 nm

3. Which type of mutation involves change in single nucleotide?
   (A) Deletion
   (B) Insertion
   (C) Point mutation
   (D) Frameshift

4. What is the hybridization of carbon in methane?
   (A) sp
   (B) sp^2
   (C) sp^3
   (D) sp^3d

5. What is the coordination number in FCC structure?
   (A) 6
   (B) 8
   (C) 12
   (D) 14

6. Which law relates induced EMF to rate of change of magnetic flux?
   (A) Lenz's law
   (B) Faraday's law
   (C) Ampere's law
   (D) Ohm's law

7. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
   (A) 25%
   (B) 33%
   (C) 50%
   (D) 75%

8. Which quantum number determines the shape of an orbital?
   (A) Principal
   (B) Azimuthal
   (C) Magnetic

(D) Spin

9. What is the escape velocity from Earth's surface?
   (A) 8 km/s
   (B) 11.2 km/s
   (C) 15 km/s
   (D) 25 km/s

10. Which reaction is endothermic?
   (A) Combustion
   (B) Neutralization
   (C) Photosynthesis
   (D) Respiration

Section B: Fill in the Blanks (1 mark each)

11. For a reversible process, entropy change of the universe is ___.
12. Raoult's law applies to ___ solutions.
13. The Krebs cycle occurs in the ___ of mitochondria.
14. The magnetic field inside a solenoid is given by B = μ■___.
15. The band gap in semiconductors is typically ___ eV.
16. The enzyme that converts prothrombin to thrombin is ___.
17. The process of splitting water during photosynthesis is called ___.
18. The uncertainty principle was given by ___.

Section C: Short Answer Questions (2 marks each)

19. Describe the mechanism of nerve impulse transmission.

20. Derive the lens formula and explain its applications.

21. Explain Le Chatelier's principle with examples.

22. Derive the expression for electric field due to a dipole.

23. Explain the structure and working of a nuclear reactor.

24. Describe electromagnetic induction and derive Faraday's law.

25. Describe the light-dependent and light-independent reactions of photosynthesis.